

13/10/25

TASK 11

CRUD Operation on graph database

AIM: TO perform CRUD operation i.e. creating, inserting, querying, finding deletion of records on graph

College management Database in SQL

1. Create Database:-

```
CREATE DATABASE COLLEGE DB;  
USE COLLEGE DB;
```

2. Create Tables:-

DEPTMENT TABLE

```
CREATE TABLE DEPARTMENT
```

```
DEPTID VARCHAR(10) PRIMARY KEY,
```

```
DEPTNAME VARCHAR(50) UNIQUE NOT NULL
```

```
);
```

STUDENT table:-

```
CREATE TABLE STUDENT
```

```
SID VARCHAR(20) PRIMARY KEY,
```

```
SNAME VARCHAR(50) NOT NULL;
```

```
AGE INT,
```

```
DEPTID VARCHAR(10),
```

```
FOREIGN KEY(DEPTID) REFERENCES
```

```
DEPARTMENT(DEPTID)
```

Shravan
[Student]

ENROLLED IN

Computer
Science
course

Rahul
[Student]

ENROLLED IN

Data
Structure
(course)



course Table

CREATE TABLE COURSE

(COURSE ID VARCHAR(10) PRIMARY KEY
COURSE NAME VARCHAR(10) NOT NULL,
CREDIT INT,
DEPT ID VARCHAR(10)
FOREIGN KEY (DEPT ID) REFERENCE
DEPARTMENT(DEPT ID)

11.

Enrollment Table:-

CREATE TABLE Enrollment

SID VARCHAR(20);
COURSE ID VARCHAR(20),
GRADE CHAR(2),

PRIMARY KEY (SID, COURSE ID),

FOREIGN KEY (SID) REFERENCES STUDENT

FOREIGN KEY (COURSE ID) REFERENCES COURSE
(COURSE ID)

3) Insert Data

Departments

Insert INTO DEPARTMENT VALUES ('D001', 'CES');

Insert INTO DEPARTMENT VALUES ('D002', 'EEE');

Insert INTO DEPARTMENT VALUES ('D003', 'MECH');

Students

Insert INTO STUDENT VALUES ('UTU14500', 'JOHN', 20, 'D001');

Insert INTO STUDENT VALUES ('UTU14501', 'ADARSH', 21, 'D002');

Insert INTO STUDENT VALUES ('UTU14502', 'VIJAY', 22, 'D001');

COURSES

```
INSERT INTO COURSE (COURSEID, COURSENAME, CREDITS)  
VALUES ('CS101', 'Data Structures', 4);  
INSERT INTO COURSE (COURSEID, COURSENAME, CREDITS)  
VALUES ('CS102', 'Algorithms', 4);  
INSERT INTO COURSE (COURSEID, COURSENAME, CREDITS)  
VALUES ('EE101', 'Electronics', 3);
```

Enrollments:-

```
INSERT INTO Enrollment VALUES ('VTU14500', 'CS101', 'A');  
INSERT INTO Enrollment VALUES ('VTU14502', 'CS102', 'B');  
INSERT INTO Enrollment VALUES ('VTU14501', 'EE101', 'A');
```

4/Queries

Select all students

SELECT * FROM Student;

Get all course taken by John

SELECT S-~~s~~name, C.courseName

FROM Student S

JOIN Enrollment E ON S.sid = E.sid

JOIN course C ON E-~~courseId~~ = C.courseId

WHERE S-~~s~~name = 'John';

Get all students in CSE Department
SELECT S.Sid, S.Sname
FROM students.

JOIN department D ON S.DeptId = D.DeptId
WHERE D.DeptName = 'CSE';

5. Update Operations

Update student Department

UPDATE student

SET DeptId = 'D001'

WHERE Sname = 'Chargana';

Update course Credits

UPDATE course

SET Credits = 5

WHERE CourseId = 'CSS101';

6. DELETE Operations

Delete a student

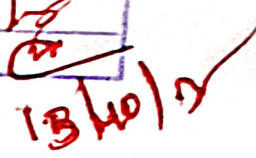
DELETE FROM student

WHERE Sname = 'Chargana';

Delete a course

DELETE FROM course

WHERE CourseId = 'CF101';

VRL - BOM - CSE	
EX NO.	1
PERFORMANCE (5)	5
RESULT ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	1
TOTAL (20)	17
SIGN WITH DATE	

13/10/22

RESULT: Thus, the ~~CRUD~~ operation in graph database using college management executed successfully.